The Box in Balance

Student sheet

Name: _____ **Date:** _____

Safety: I hold the box in both hands, over the table. The cable stays slack. I never throw the box to a classmate: I put it down on the table.

1. What I see on the screen

On the screen you can see three numbers: **AX**, **AY** and **AZ**. The box feels whatever holds it up **pushing it upwards** so that it does not fall — the table, or your hand. The numbers show how big that push is in each of the three directions.

The activity starts with the **Start** button. You move on with the green button at the bottom right, you go back with **Back**, and at the last step the green button reads **Finish**.



2. What I do on my own

First I predict. I do not change my prediction after I measure.

1. The box is lying still on the table. I think AZ will show:
☐ zero ☐ something other than zero if you can, write the number too: _____

because _____

_____

2. When I tilt the box to the right, I think AX will:
☐ go up ☐ go down ☐ stay the same

because _____

_____

Now I measure. I put the box down, wait for it to be still, count to three, then write the numbers exactly as they appear on the screen.

Position of the box	AX	AY	AZ
Flat on the table			
Tilted to the right			
Tilted to the left			
Tipped forward			
On its edge			
Upside down			
Flat, a second time			

When I tilt the box, the number AX: ☐ ↑ ☐ ↓ ☐ →
(tick one: goes up, goes down, stays the same)

Challenge. Can you make all three numbers zero at the same time? ☐ YES ☐ NO

3. The conclusion I write

I predicted that _____

_____

The data show that _____

_____

The difference is explained by _____

_____

Write one sentence: why the box does not show zero when it is sitting still.

_____

Exit ticket

Write one thing that surprised you today.

_____